

Proof. It is sufficient to apply Lemma 1.8 to the following setting: $X = Y = \mathbb{R}^k$, $\tilde{X} = \tilde{Y} = \mathcal{P}(\Omega)$, where $\Omega \subset \mathbb{R}^{d-k}$ is such that $\text{spt}(\mu), \text{spt}(\nu) \subset \mathbb{R}^k \times \Omega$. The space $\mathcal{P}(\Omega)$ is endowed with any distance metrizing weak convergence and making this space a separable metric space (see, for instance, Chapter 5 for a choice of distances with this property, or build it with standard methods for the weak convergence in the unit ball of a Banach space). For the cost function which appears in the statement of Lemma 1.8 we take $(a, b) \mapsto \mathcal{T}_c(a, b)$ and $a(x^+) := \mu^{x^+}$, $b(y^+) := \nu^{y^+}$. Since this quantity is bounded by $\max |c|$ (as we are on compact supports), there is no difficulty in choosing two bounded functions f, g so that $\mathcal{T}_c(a, b) \leq f(a) + g(b)$. $\square$ $\square$

Another easy lemma that we need is the following.

Lemma 2.25. *Given $\mu \in \mathcal{P}(X)$ and $\nu_n \in \mathcal{P}(\Omega)$, where $\Omega \subset \mathbb{R}^d$ is a compact set, suppose that $\gamma_n := \gamma_{T_n} \in \Pi(\mu, \nu_n)$ is such that $\gamma_n \rightarrow \gamma = \gamma_T$. Then $T_n \rightarrow T$ in $L^2(X, \mu)$.*

Proof. Since the functions T_n are valued in Ω , they are equibounded and they admit a weak limit in $L^2(X, \mu)$ up to subsequences. Let us call S this limit. To prove $S = T$, take a continuous function $\xi \in C(X; \mathbb{R}^d)$ and consider the limit of

$$\begin{aligned} \int (\xi(x) \cdot S(x)) d\mu(x) &= \lim_n \int (\xi(x) \cdot T_n(x)) d\mu(x) = \lim_n \int (\xi(x) \cdot y) d\gamma_n(x, y) \\ &= \int (\xi(x) \cdot y) d\gamma_T(x, y) = \int (\xi(x) \cdot T(x)) d\mu(x). \end{aligned}$$

This proves $S = T$.

In order to prove strong convergence, just consider

$$\int |T_n|^2 d\mu = \int |y|^2 d\gamma_n \rightarrow \int |y|^2 d\gamma = \int |T|^2 d\mu.$$

This proves that we do not only have weak convergence in L^2 , but also convergence of the L^2 norm. This implies strong convergence, as usual in Hilbert spaces. $\square$ $\square$

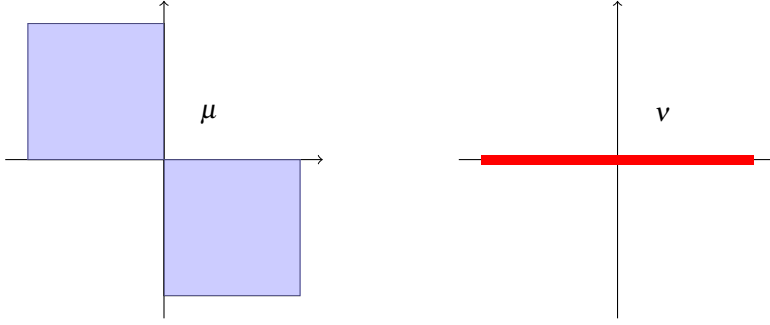
Note that the above proof also works in other spaces L^p and that the boundedness assumption (i.e. compact supports) could be replaced by suitable integrability bounds.

Let us remark here that if instead of considering the quadratic cost c_ε , one considers the more general separable cost

$$c_\varepsilon(x, y) := \sum_{i=1}^d \lambda_i(\varepsilon) c_i(x_i - y_i)$$

where each c_i is a smooth strictly convex function (with suitable growth), then the previous convergence proof carries over.

Example 2.26 (A counterexample when the measures have atoms). We now show that interestingly, the absolute continuity hypothesis in Theorem 2.23 is necessary not only for μ , but also for ν (see Remark 2.21 to check the definition of the Knothe map in such a case). We propose a very simple example in $\mathbb{R}^2$ where μ is absolutely continuous

Figure 2.3: The measures μ and ν in the counter-example

with respect to the Lebesgue measure but ν is not, and we show that the conclusion of the Theorem fails. On the square $\Omega := [-1, 1] \times [-1, 1]$, define $\mu = \frac{1}{2} \mathbb{1}_E \cdot \mathcal{L}^2$, where $E = \{(x_1, x_2) \in [-1, 1] \times [-1, 1] : x_1 x_2 < 0\}$, so that the measure μ is uniformly spread on the upper left and the lower right quadrants, and $\nu = \frac{1}{2} \mathcal{H}_S^1$, S being the segment $[-1, 1] \times \{0\}$.

The Knothe-Rosenblatt map is easily computed as

$$T(x) := (2x_1 + \text{sign}(x_2)), 0)$$

(indeed, the measure ν^2 is a Dirac mass at 0, hence $T^2(x) = 0$). The solution of any transportation problem with cost $|x_1 - y_1|^2 + \varepsilon |x_2 - y_2|^2$ is $T^0(x) := (x_1, 0)$ (no transport may do better than this one, which projects on the support of ν). Therefore, in this example the optimal transportation maps fail to tend to the Knothe-Rosenblatt map. The reason is the atom in the measure $\nu^2 = \delta_0$.

2.5 Discussion

2.5.1 Histogram equalization

A very typical problem where we encounter monotone transport maps in image processing is histogram equalization. The setting is easy to explain: take a black and white digital picture, and consider all possible grey levels; these levels are number from 0 to 255; for each $i = 0, \dots, 255$ count how many pixels have exactly such a grey level, and make an histogram of that. In some situations, it could be interesting to transform the given picture into a different one, which should represent again the same subject, but have a different histogram. Why? a typical example is the following: you have two pictures, A and B, which are supposed to represent the same object, with minor modifications (something has moved. . .), at different instants of time; you want to compare them, but B happens to be much darker than A (this means that the histogram of B is much more concentrated on the right-hand part of the graph than that of A). Then, before comparing them, one prefers to transform B into a new picture B', where the darkness level of B' is comparable to that of A, i.e. the histogram of B' is equal to that

of A. Similarly, one can also consider the case where A and B represent two parts of a same object, and the two pictures must be merged, but it could happen that the two partial pictures have been taken from different angles, and the light is different.

In some other cases we have only one picture, but its histogram is too much concentrated around one peak. This means that pixels have different grey levels, but not so different. In order to see better what is inside the picture, one would like to enhance the contrast, which corresponds to using a more widespread histogram. In such a case one fixes a target histogram, for instance the uniform one, where every level is used by the same number of pixels. This is the best possible histograms in the sense that we do not waste information by concentrating too much around a same grey level.

In all these cases, we want to perform a transformation which does not act pixel by pixel, but grey level by grey level: we look for a map T from the set X of grey levels to itself (all pixels with grey level x will be turned into the grey level $T(x)$), and we want to prescribe the histogram of the image after the transformation. This is the same as saying that we have a measure μ on X and we want to impose a condition of the form $T_{\#}\mu = \nu$. Here the important fact is that the space X is a subset of $\mathbb{R}$, i.e. it is one-dimensional and ordered. The natural condition on T is that it should be monotone increasing. This condition translates the fact that we do not want to invert darker and brighter pixels.

Up to the fact that, in this discrete setting (the grey level only takes value in a finite set of integers), the two measures μ and ν are atomic (and hence a transport map could not exist, the situation is exactly the one described in this chapter. The problem of non-existence of transport maps could be solved in several ways in practice: one could introduce some rounding rules, or use a transport plan, which corresponds to use different colors on pixels which had the same colors on the original picture, i.e. splitting the mass.

The situation becomes trickier when we take color images. In this case the space X of colors becomes three-dimensional, and the notion of monotone transport does not make sense any more. Obviously the quadratic optimal transport (Brenier's map) could play the role of a monotone map, since it is monotone in the vector sense $(T(x) - T(y)) \cdot (x - y) \geq 0$. On the other hand, it is expensive to compute (we will see in Chapter 6 some numerical methods to do it, and the associated difficulties) and some of its features are probably useless for this kind of application in image processing (the fact that it is a gradient, for instance).

Several solutions are now matter of current research: looking for transport maps which are more regular but not so far from the optimal one ([164]), or producing maps defined from the composition/superposition of several one-dimensional bricks, so that they keep some monotonicity properties and are easier to compute (see [252] and Section 2.5.2).

Also, in some cases all the components of the color space are not equally important, and one could decide to look only at one or two components, or to treat them separately. In this case, colors are often described by different sets of parameters.

See, for instance, [141], where some applications of a 1D analysis on the circle are proposed for *hue* component of the color space (with the HSL decomposition into Hue, Saturation, and Lightness).

The same space of colors can also be written as the combination of a 1D informa-

tion (the *luma* signal, corresponding to a weighted average of the three components R, G and B, and standing for the brightness of the image) and of a 2D information (called *chrominance* or *chroma*, standing for the color information). In Figure 2.4 we see an application of optimal transport in 1D and 2D to the color transfer between a two streets with very different colors.

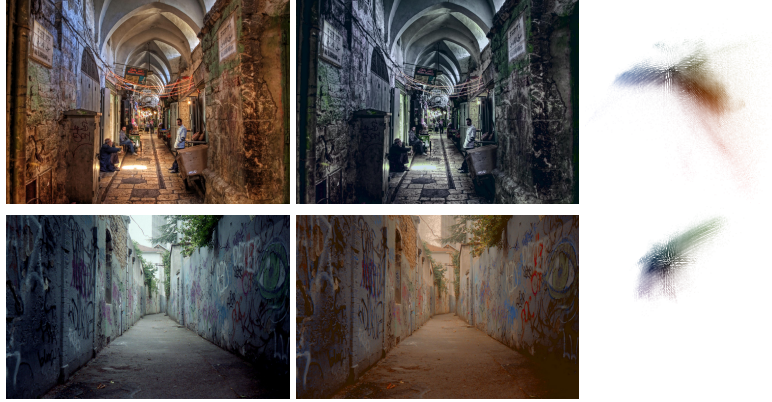


Figure 2.4: First column: the images of a souq arcade and of a narrow Western street with tags on the walls, in different lights. Second column: the two images, each with the colors of the other. Third column: their chrominance histograms (in the CIE-lab representation). The transfer has been done independently on the 1D feature of the luma on the 2D feature of the chroma. Only the chroma histograms are represented. Pictures kindly provided by G. Peyré.

For a wider discussion about 1D histogram equalization and 3D color transfer in connection with optimal transport, we refer to [164] and [255], and the references therein (in particular some older papers such as [259, 242]).

2.5.2 Efficient monotone maps from 1D constructions

For many applications, in particular in image processing (as in the previous Section 2.5.1), people tried to build “good” transport maps, which are not necessarily optimal, but satisfy at least two criteria: they must be computationally easy to find, and they must have some “monotonicity” properties. A good idea to cope with these two requirements is to build these maps by using one-dimensional bricks.

A very first idea that one could have is to use the Knothe map. Yet, the main drawback of such a choice is that it is highly anisotropic. The choice of the orthonormal basis is crucial, and the map gives very different roles to the different axes.

A different idea was contained in [252], where it is proposed the so-called *Iterative Distribution Transfer algorithm*: a source measure μ is given, together with a target measure ν ; starting from $\mu_0 = \mu$ the authors build recursively a sequence of maps T_n and set $\mu_{n+1} = (T_n)_\# \mu_n$. The sequence is built so that μ_n should converge to ν , which

Box 2.4. – Good to know! – The Radon or X-ray transform

Definition - Given a measure μ on $\mathbb{R}^d$, we define the Radon transform $R\mu$ of μ as the family of measures (μ_e) , parametrized by $e \in \mathbb{S}^{d-1}$, and defined as $(\pi_e)_\# \mu$, where $\pi_e(x) := x \cdot e$. When μ is absolutely continuous, these measures are also absolutely continuous, and we can define $R\mu$ as a function over $\mathbb{S}^{d-1} \times \mathbb{R}$, through $R\mu(e, t) = f_e(t)$ and $\mu_e = f_e(t)dt$. In dimension $d > 2$, one could consider projections onto hyperplanes instead of lines (note that projecting onto all $(d-1)$ -dimensional hyperplanes means considering the integral of the measure (or of its density) on arbitrary one-dimensional lines. This transformation is called X-ray transform, but Radon and X-ray transforms coincide for $d = 2$. See for instance [193] for more details and applications.

Proposition - Given two measures μ, ν on $\mathbb{R}^d$, if their Radon transforms coincide, then $\mu = \nu$.

Proof - Note that $\int_{\mathbb{R}^d} e^{i\xi \cdot x} d\mu = \int_{\mathbb{R}} e^{it|\xi|} d\mu_{\xi/|\xi|}(t) dt$. This means that the Radon transform uniquely determines the Fourier transform of a measure, so that $R\mu = R\nu \Rightarrow \hat{\mu} = \hat{\nu}$. The equality $\mu = \nu$ follows from standard results about the Fourier transform.

The Radon and the X-ray transform are very well known objects in applications such as tomography. Computer tomography devices in medical images are exactly based on the inversion of the X-ray transform.

means that, for large n , the map $T_n \circ T_{n-1} \circ \dots \circ T_1 \circ T_0$ is a transport map from μ to a measure μ_n very close to ν .

In the construction of [252] the maps T_n are obtained in the following way: at every step n an orthonormal basis B_n is randomly selected; then, we define T_n^j as the monotone transport map from $(\pi_j)_\# \mu_n$ to $(\pi_j)_\# \nu$, where π_j is the projection on the j -th coordinate of the selected basis; the map T_n is defined, in the corresponding basis, as $T_n(x) = (T_n^1(x_1), \dots, T_n^d(x_d))$.

There are two ideas behind this choice: first, all the maps T_n are easy to compute, because composed of finitely many 1D monotone maps, and have monotonicity properties; second, if the basis B_n are chosen in a suitable diversified way, then the only possibility for the algorithm to stop on a constant measure μ_n is that μ_n and ν have the same projections on all directions. This implies $\mu_n = \nu$, thanks to the considerations on Radon and X-ray transforms of Box 2.4, and obviously suggests that the limit of μ_n should be ν .

Yet, this convergence is observed empirically but proven in [252] only if ν is Gaussian and the bases B_n are i.i.d. random variable, uniform on the set of orthonormal bases. See also Chapter 5 in [66] for a revised and more general proof.

Another interesting idea on how to build similar transport maps, is due to M. Bernot. Many features are in common with the approach of [252].

Consider two measures $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ and project them onto any one-dimensional direction. For every $e \in \mathbb{S}^{d-1}$ (the unit sphere of $\mathbb{R}^d$), we take the map $\pi_e : \mathbb{R}^d \rightarrow \mathbb{R}$ given by $\pi_e(x) = x \cdot e$ and look at the image measures $(\pi_e)_\# \mu$ and $(\pi_e)_\# \nu$. Again, they are measures on the real line, and we call $T_e : \mathbb{R} \rightarrow \mathbb{R}$ the monotone optimal transport between them. The idea is that, as far as the direction e is concerned, every point x of $\mathbb{R}^d$ should be displaced of a vector $S_e(x) := (T_e(\pi_e(x)) - \pi_e(x))e$. To do a global dis-



Figure 2.5: Optimal transport between a finite number of points on an annulus and on a duck. Colors represent the assignment of points in the annulus to points on the duck. Middle: 1D-sliced assignment; right: optimal map for the quadratic cost. The two results are very similar (but one can find few points with different colors, which means a slightly different assignment). Picture taken from [256] with permission.

placement, consider $S(x) = \int_{\mathbb{S}^{d-1}} S_e(x) d\mathcal{H}^{d-1}(e)$, where $\mathcal{H}^{d-1}$ is the uniform measure on the sphere.

Again, there is no reason to guarantee that $\text{id} + S$ is a transport map from μ to ν . But if one fixes a small time step $\tau > 0$ and uses a displacement τS getting a measure $\mu_1 = (\text{id} + \tau S)_\# \mu$, then it is possible to iterate the construction, as it is done in [252]. Only the definition of the transport map T_n is different. Again, one expects the sequence of measures μ_n that are built in this way to converge to ν , but the problem is not really solved. From the empirical point of view, the transport maps that are obtained in this way are quite satisfactory, and have been tested in particular in the discrete case (a finite number of Dirac masses with equal mass, i.e. the so-called *assignment problem*). An example is given in Figure 2.5.2, where there is a comparison between the optimal transport obtained via linear programming methods (see Section 6.4.1) and the assignment obtained via these 1D-sliced ideas by M. Bernot.

Moreover, we will see in Section 5.5 that one can also build an interesting distance on the set of probabilities with this kind of idea (taking the average of a distance defined direction by direction: it will be much easier to compute than the corresponding multidimensional distance). In Section 8.4 we will also show that the iterative algorithm above corresponds indeed to a gradient flow evolution.

2.5.3 Isoperimetric inequality via Knothe or Brenier maps

For the last discussion section of this chapter we review one of the most spectacular applications of optimal transport, i.e. its role in geometric inequalities. We will see here how the isoperimetric inequality can be proven either by using the optimal (Brenier) map, or even the Knothe map.

It is interesting to learn that it was indeed one of the first motivations for the use of Knothe transport to deal with geometric inequalities (see [204]). However, the short proof of the isoperimetric inequality that we see here was first found by Gromov (see the appendix in [236]), while the original goal of Knothe in [204] was to establish other

inequalities concerning convex sets.

The isoperimetric inequality states that every set E has a larger perimeter than the ball B with the same volume $|B| = |E|$. Thanks to the scaling properties of the volume and the perimeter, it is possible to write this fact in the following way:

$$\text{Per}(E) \geq d\omega_d^{1/d}|E|^{1-1/d},$$

where ω_d is the volume of the unit ball in $\mathbb{R}^d$.

As usual in these discussion sections, we will not give full details on the ideas that we mention, and we will be quite sloppy concerning regularity. Anyway, think that E is a smooth domain in $\mathbb{R}^d$. By scaling, we can also suppose that E has the same volume of the unit ball $B = B(0, 1)$

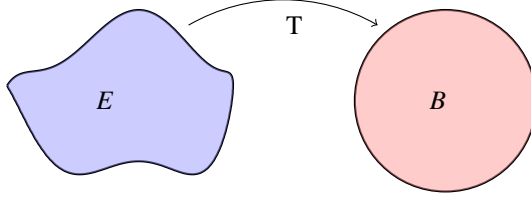
The idea to prove this result is the following: consider the densities $\mu = \mathcal{L}_E$ and $\nu = \mathcal{L}_B$, and use the Knothe transport between these densities, denoted by T . Due to the considerations on the Jacobian matrix of T , it has only positive eigenvalues and it satisfies the condition

$$\det(DT) = \frac{|B|}{|E|} = 1$$

(with no absolute value at the determinant). Hence we can write

$$\omega_d = |B| = |E| = \int_E (\det(DT))^{1/d} \leq \frac{1}{d} \int_E \nabla \cdot T = \frac{1}{d} \int_{\partial E} T \cdot \mathbf{n} \leq \frac{1}{d} \text{Per}(E), \quad (2.8)$$

where the inequalities are obtained thanks to the arithmetic-geometric inequality $\frac{1}{d} \sum_i \lambda_i \geq (\lambda_1 \dots \lambda_d)^{1/d}$ applied to the eigenvalues of DT (which gives an inequality between the determinant and the trace, i.e. the divergence of T), and to the fact that T is valued in B , whence $T \cdot \mathbf{n} \leq 1$. This shows $\text{Per}(E) \geq d\omega_d = \text{Per}(B)$. Note that we only used the



fact that the Jacobian matrix of T has positive eigenvalues and that it transports one measure onto the other. These properties are also satisfied by the Brenier map, simply this map was not known at the time this proof was first performed.

The same proof can also be used to prove the uniqueness of the optimal sets (i.e. if a set E is such that $\text{Per}(E) = d\omega_d$, then it is a unit ball, up to negligible sets).

The use of the Brenier map strongly simplifies the study of the equality cases. Suppose indeed to have equality in the above inequalities. This implies the equality $(\det(DT))^{1/d} = (\nabla \cdot T)/d$ a.e. and, from the cases of equality in the arithmetic-geometric inequality, we get that the eigenvalues of DT should all be equal. Thanks to the condition on the determinant, they are equal to 1. If the matrix is known to be symmetric

(which is the case for Brenier's map, but not for Knothe's one), then we get $DT = I$ and $T(x) = x + x_0$. Since T would be a translation, then we conclude $E = B$, up to translations.

It is slightly harder for the Knothe transport, since DT is not symmetric. Yet, if all the eigenvalues are 1, one deduces that T^d (the last component) is a translation. If we suppose from the beginning that (up to a fixed translation) E and B share the same barycenter, then this translation is the identity. This means $(\pi_d)_\# \mu = (\pi_d)_\# \nu$. This is not yet enough to get $\mu = \nu$, and hence $E = B$, unless one uses the fact that the Knothe transport is not isotropic (i.e. its definition depends on the choice of the axes), and that the directions of the axes are arbitrary. Hence, if E is a set giving equality in the isoperimetric inequality, then its projections onto any direction coincide with those of the ball. This implies $E \subset B$ and, by equality of the volumes, $E = B$ (the argument may also be done through the measures and the notion of Radon transform, since $(\pi_d)_\# \mu = (\pi_d)_\# \nu$ for arbitrary direction of the d -th axis implies $\mu = \nu$).

We finish the section by stressing that, if the isoperimetric inequality can obviously be proven in many other ways, one recent advanced refinement of it (i.e. the quantitative version of the anisotropic perimetric inequality, where the definition of perimeter is modified, and the optimal set is no more the ball but another convex set, unit ball of another norm) was only proven with the optimal exponent via optimal transport techniques (see [169], and [274] for a short review).

Chapter 3

L^1 and L^∞ theory

Chapter 1 gave, in the framework of the general theory of optimal transportation based on duality methods, an existence result for the optimal transport map when the cost is of the form $c(x, y) = h(x - y)$ and h is strictly convex. In particular, using $c(x, y) = |x - y|^p$, this applies to the minimization problems

$$\min \{ \|T - \text{id}\|_{L^p(\mu)} : T_{\#}\mu = \nu \},$$

(expressed in Kantorovich terms as $\min \{ \|y - x\|_{L^p(\gamma)} : \gamma \in \Pi(\mu, \nu) \}$), for $p \in]1, +\infty[$. We look in this chapter at the two limit cases $p = 1$ and $p = \infty$, which require additional techniques.

Then, the discussion section will go into two different directions: on the one hand the L^1 and L^∞ cases introduced and motivated the study of convex costs which could be non strictly-convex or infinite-valued somewhere, and the last developments on this topic will be debated in Section 3.3.1; on the other hand one could wonder what is the situation for $0 < p < 1$, and more generally for costs which are concave increasing functions of the distance, which will be the subject of Section 3.3.2.

3.1 The Monge case, with cost $|x - y|$

This section will prove that, if $\mu, \nu \in \mathcal{P}(\Omega)$, where $\Omega \subset \mathbb{R}^d$ is a compact domain, and $\mu \ll \mathcal{L}^d$, then there exists an optimal transport map T for the cost $|x - y|$ (the Euclidean distance in $\mathbb{R}^d$). The proof that we present is essentially due to Ambrosio [8], as a modification from that of Sudakov, [290]. In the introduction we gave references for the other alternative proofs of the existence in this case, and in the discussion section we will see their generalizations.

3.1.1 Duality for distance costs

Let us spend some words on the case where the cost $c(x, y)$ is actually a distance (thus satisfying the triangle inequality, vanishing if $x = y \dots$). Since the cost is symmetric, we will avoid the distinction between the $\bar{c}$ and the c transform.